



### "Will come to a bad end"

Berkshire Hathaway CEO, Warren Buffett believes that cryptocurrencies will come to a bad end.  
<http://dgit.in/BadCrypto>



### Europe prepared for AI job losses

Intel AI boss, Naveen Rao believes that Europeans due to better access to education are better prepared for AI-related job losses. <http://dgit.in/EuroAI>

With Firebase Realtime Database on the Blaze pricing plan, you can support your app's data needs at scale by splitting your data across multiple database instances in the same Firebase project. Stream-line authentication with Firebase Authentication on your project and authenticate users across your database instances. Control access to the data in each database with custom Firebase Realtime Database Rules for each database instance.

#### How does it work?

The Firebase Realtime Database lets you build rich, collaborative applications with client side code being able to securely and directly access the database. Local data persistence is maintained, and realtime events continue to fire even while offline, giving the end user a responsive experience. When the device regains connection, the Realtime Database synchronizes the local data changes with the remote updates that occurred while the client was offline, merging any conflicts automatically.

The Realtime Database provides a flexible, expression-based rules language, called Firebase Realtime Database Security Rules, to define how your data should be structured and when data can be read from or written to. When integrated with Firebase Authentication, developers can define who has access to what data, and how they can access it.

The Realtime Database is a NoSQL database and as such has different optimizations and functionality compared to a relational database. The Realtime Database API is designed to only allow operations that can be executed quickly. This enables you to build a great realtime experience that can serve millions of users without compromising on responsiveness.

#### How to implement?

- **Integrate the Firebase Realtime Database SDKs** - Quickly include clients via Gradle, CocoaPods, or a script include.

- **Create Realtime Database References** - Reference your JSON data, such as `users/user:1234/phone_number` to set data or subscribe to data changes.
- **Set Data and Listen for Changes** - Use these references to write data or subscribe to changes.
- **Enable Offline Persistence** - Allow data to be written to the device's local disk so it can be available offline.
- **Secure your data** - Use Firebase Realtime Database Security Rules to secure your data.

### CLOUD STORAGE

Cloud Storage for Firebase is a powerful, simple, and cost-effective object storage service built for Google scale. The Firebase SDKs for Cloud Storage add Google security to file uploads and downloads for your Firebase apps, regardless of network quality. You can use these SDKs to store images, audio, video, or other user-generated content. On the server, you can use Google Cloud Storage, to access the same files.

#### Features

- **Robust operations** - Firebase SDKs for Cloud Storage perform uploads and downloads regardless of network quality. Uploads and downloads are robust, meaning they restart where they stopped.
- **Strong security** - Firebase SDKs for Cloud Storage integrate with Firebase Authentication. You can use the declarative security model to allow access based on filename, size, content type, and other metadata.

- **High scalability** - Cloud Storage for Firebase is built for exabyte scale when your app grows in popularity.

#### How does it work?

Developers use the Firebase SDKs for Cloud Storage to upload and download files directly from clients. If the network connection is poor, the client is able to retry the operation right where it left off, saving your users time and bandwidth.

Cloud Storage stores your files in a Google Cloud Storage bucket, making them accessible through both Firebase and Google Cloud. This allows you the flexibility to upload and download files from mobile clients via the Firebase SDKs and do server-side processing such as image filtering or video transcoding using Google Cloud Platform. Cloud Storage scales automatically, meaning that there's no need to migrate to any other provider. Learn more about all the benefits of the integration with Google Cloud Platform.

The Firebase SDKs for Cloud Storage integrate seamlessly with Firebase Authentication to identify users, and we provide a declarative security language that lets you set access controls on individual files or groups of files, so you can make files as public or private as you want.

#### How to implement?

- **Integrate the Firebase SDKs for Cloud Storage** - Quickly include clients via Gradle, CocoaPods, or a script include.
- **Create a Reference** - Reference

```
// Read from the database
myRef.addValueEventListener(new ValueEventListener() {
    @Override
    public void onDataChange(DataSnapshot dataSnapshot) {
        // This method is called once with the initial value and again
        // whenever data at this location is updated.
        String value = dataSnapshot.getValue(String.class);
        Log.d(TAG, "Value is: " + value);
    }

    @Override
    public void onCancelled(DatabaseError error) {
        // Failed to read value
        Log.w(TAG, "Failed to read value.", error.toException());
    }
});
```

Read from database